Experiment-2

Aim:

To convert the ER diagram of an Indian e-commerce platform into a relational database schema and implement it using MySQL with primary keys, foreign keys, NOT NULL, UNIQUE and DELETE CASCADE/SET NULL constraints. To insert sample data and demonstrate referential integrity violations.

Source code:

```
CREATE DATABASE ECommerceDB;
USE ECommerceDB;
--1. ADDRESS
CREATE TABLE Address(
    AddressID INT PRIMARY KEY,
    HouseNo VARCHAR(20) NOT NULL,
    Street VARCHAR(100) NOT NULL,
    City VARCHAR(50) NOT NULL,
    State VARCHAR(50) NOT NULL,
    Pincode VARCHAR(10) NOT NULL
);
--2. CUSTOMER
CREATE TABLE Customer(
    CustomerID INT PRIMARY KEY,
    CustomerName VARCHAR(100) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    AddressID INT,
    FOREIGN KEY (AddressID)
      REFERENCES Address(AddressID)
      ON DELETE SET NULL
);
```

```
--3. Customer Phone table
CREATE TABLE CustomerPhone(
    CustomerID INT,
    PhoneNo VARCHAR(15) NOT NULL,
    PRIMARY KEY (CustomerID, PhoneNo),
    FOREIGN KEY (CustomerID)
        REFERENCES Customer (CustomerID)
        ON DELETE CASCADE
);
--4. Category Table
CREATE TABLE Category(
    CategoryID INT PRIMARY KEY,
    CategoryName VARCHAR(50) NOT NULL UNIQUE
);
--5. Product Table
CREATE TABLE Product(
    ProductID INT PRIMARY KEY,
    ProductName VARCHAR(100) NOT NULL,
    Price Decimal (10,2) NOT NULL,
    StockQuantity INT NOT NULL,
    Description VARCHAR(255),
    CategoryID INT NOT NULL,
    FOREIGN KEY (CategoryID)
        REFERENCES category(CategoryID)
        ON DELETE CASCADE,
    CHECK (Price >= 0),
    CHECK (StockQuantity >= 0)
);
--6. Physical Product
CREATE TABLE PhysicalProduct(
    ProductID INT PRIMARY KEY,
    Weight DECIMAL(8,2),
    WarrantyMonths INT,
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    FOREIGN KEY (ProductID)
        REFERENCES Product (ProductID)
        ON DELETE CASCADE
);
--7. Digital Product
CREATE TABLE DigitalProduct(
    ProductID INT PRIMARY KEY,
    FileSizeMB DECIMAL (10,2),
    DownloadURL VARCHAR (255),
    FOREIGN KEY (ProductID)
        REFERENCES Product (ProductID)
        ON DELETE CASCADE
);
--8. SELLER TABLE
CREATE TABLE Seller(
    SellerID INT PRIMARY KEY,
    SellerName VARCHAR (100) NOT NULL,
    Email VARCHAR (100) NOT NULL UNIQUE,
    PhoneNo VARCHAR (15),
    GSTIN VARCHAR (20) UNIQUE
);
--9. Seller_Product Table
CREATE TABLE SellerProduct(
    SellerID INT,
    Product INT,
    SellingPrice DECIMAL (10,2) NOT NULL,
    PRIMARY KEY (SellerID, ProductID),
    FOREIGN KEY (SellerID)
        REFERENCES Seller (SellerID)
        ON DELETE CASCADE,
    FOREIGN KEY (ProductID)
        REFERENCES Product (ProductID)
        ON DELETE CASCADE,
CHECK (SellingPrice >= 0)
);
```

```
--10. Order Table
CREATE TABLE Orders(
    OrderID INT PRIMARY KEY,
    OrderDate DATE NOT NULL,
    OrderStatus VARCHAR(30) NOT NULL,
    TotalAmount DECIMAL(10,2) NOT NULL,
    CustomerID INT NOT NULL,
    FOREIGN KEY (CustomerID)
      REFERENCES Customers(CustomerID)
      ON DELETE CASCADE,
    CHECK(TotalAmount >= 0)
);
--11. Order Item Table
CREATE TABLE OrderItem(
    OrderID INT,
    ProductID INT,
    Quantity INT NOT NULL,
    UnitPrice DECIMAL(10,2) NOT NULL,
    PRIMARY KEY (OrderID, ProductID),
    FOREIGN KEY (OrderID)
      REFERENCES Orders(OrderID)
      ON DELETE CASCADE,
    FOREIGN KEY (ProductID)
      REFERENCES Product(ProductID)
      ON DELETE CASCADE,
    CHECK(Quantity > 0),
    CHECK(UnitPrice >= 0)
);
-- 12. Payment Table
CREATE TABLE Payment(
    PaymentID INT PRIMARY KEY,
    PaymentDate DATE NOT NULL,
    Amount DECIMAL(10,2) NOT NULL,
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    PaymentMethod VARCHAR(30) NOT NULL,
    PaymentStatus VARCHAR(30) NOT NULL,
    OrderID INT NOT NULL UNIQUE,
    FOREIGN KEY (OrderID)
      REFERENCES Orders(OrderID)
      ON DELETE CASCADE,
    CHECK (Amount >= 0)
);
--13. Delivery Table
CREATE TABLE Delivery(
    DeliveryID INT PRIMARY KEY,
    DeliveryDate DATE,
    DeliveryStatus VARCHAR(30) NOT NULL,
    TrackingNumber VARCHAR(50) UNIQUE,
    OrderID INT NOT NULL UNIQUE,
    FOREIGN KEY(OrderID)
      REFERENCES Orders(OrderID)
      ON DELETE CASCADE
);
-- Insert Sample data.
INSERT INTO Address
    VALUES (101, '12A', 'Hazratganj', 'Lucknow', 'UttarPradesh',
            '226001');

INSERT INTO Customer
VALUES (1, 'abc', 'abc@gmail.com', 101);

INSERT INTO CustomerPhone
VALUES (1, '9876543210');

INSERT INTO Category
VALUES (1, 'Electronics');

INSERT INTO Product
```

```
JOIN Product p
    ON oi.ProductID = p.ProductID
JOIN Category cat
    ON p.CategoryID = cat.CategoryID
JOIN Payment pay
    ON o.OrderID = pay.OrderID
JOIN Delivery d
    ON o.OrderID = d.OrderID;
```

Output

OrderID	CustomerName	ProductName	Category
5001	abc	Laptop	Electronics
5002	abc	T-Shirt	Clothing

UnitPrice	OrderStatus	PaymentMethod	DeliveryStatus
50000.00	Delivered	UPI	Delivered
1200.00	Delivered	UPI	Delivered

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```
VALUES (101, 'Laptop', 55000.00, 20, 'Student laptop', 1);
INSERT INTO PhysicalProduct
VALUES (101, 1.80, 24);
INSERT INTO DigitalProduct
VALUES(104, 850.50, 'https://example.com/course');
INSERT INTO Seller
VALUES (201, 'TechWorld India', 'techworld@gmail.com',
        '9876500001', '09ABCDE1234F1Z5');
INSERT INTO SellerProduct
VALUES (201, 101, 5400 000);
INSERT INTO Orders
VALUES (5001, '2026-08-10', 'Delivered', 54700.00, 1)
INSERT INTO OrderItem
VALUES (5001, 101, 1, 54000.00);
INSERT INTO Payment
VALUES (9001, '2026-08-10', 54700.00, 'UPI', 'Paid', 5001);
INSERT INTO Delivery
VALUES (7001, '2026-08-13', 'Delivered', 'TRK10001', 5001);

SELECT
    o.OrderID,
    c.CustomerName,
    p.ProductName,
    cat.CategoryName,
    oi.Quantity,
    oi.UnitPrice,
    o.OrderStatus,
    pay.PaymentMethod,
    d.DeliveryStatus
FROM Orders o
JOIN Customers c
  ON o.CustomerID = c.CustomerID
JOIN OrderItem oi
  ON o.orderID = oi.OrderID
```